

targeted scenario generation
to close gaps

Test execution across the ODD (PEGASUS three pillars) → Datasets

Simulation

CARLA 0.9.15

6 maps · 2,050 scenes
used in this study

Proving Ground

Proving-ground logs

not used in this study

Public Road

Argoverse 2.0

250k scenarios · 17k used
used in this study

**Traffic Scene Graph
Construction**

map graph G_{map} +
actor graph G_{actor}
→ hierarchical
scene graph G

This paper's contribution

Subgraph Isomorphism

18 archetypes (VF2)

**GINE Graph
Embeddings**

contrastive (NT-Xent)

Coverage Argument

structural & parameter-level
coverage gaps

**Safety Case / Re-
lease Argument**

ISO 21448 (SOTIF), UL 4600
*outside this paper's scope—
safety, not completeness*